

Market Details

Build an AI Prior Authorization + Denial Appeal Copilot for healthcare providers

This is the strongest mix of:

- huge market,
- painful workflow,
- clear ROI,
- real AI,
- and realistic buildability for a course project.

Why this one: U.S. health spending hit **\$5.3 trillion in 2024**, and prior authorization is still a massive friction point. In Medicare Advantage alone, insurers processed **52.8 million** prior-authorization determinations in 2024; **4.1 million** were denied, only **11.5%** of denials were appealed, and **80.7%** of appealed denials were overturned. CMS has also been actively pushing technology changes to improve prior authorization and reduce burden, which is a strong signal that this pain is real and current. ([Centers for Medicare & Medicaid Services](#))

This also lines up extremely well with the a16z themes in your uploaded notes: Jennifer Li's "messy multimodal data," Kimberly Tan's view that reasoning, multimodality, and computer use unlock net-new companies, including **billing disputes** and claims-style adjudication, and David Haber's point that the best AI startups improve the customer's business model, not just automate a random task. Your notes also explicitly point to AI-native banking and insurance infrastructure as a major opportunity area.

What the product would do

A clinic or hospital staff member uploads:

- doctor notes,
- lab results,

- diagnosis codes,
- insurer policy PDFs,
- denial letters,
- and treatment requests.

Your system then:

- detects whether prior authorization is likely required,
- extracts the relevant clinical facts,
- matches them against payer policy rules,
- identifies missing evidence,
- predicts denial risk,
- drafts the submission packet,
- and, if denied, drafts an appeal letter with supporting rationale.

That is not a toy chatbot. That is a real AI workflow product.

Why this can make a lot of money

Because buyers do not pay for “AI.” They pay for:

- fewer denials,
- faster approvals,
- less staff time,
- faster cash flow,
- and fewer delayed treatments.

This is exactly the kind of workflow where software can become **revenue-accrative**, not just “nice to have.” If a clinic gets more approvals and fewer delays, your product directly protects or unlocks revenue. That matches the a16z idea that the best AI startups reinforce the customer’s business model itself.

Also, the burden is bad enough that many practices dedicate staff to it. AMA says prior auth creates substantial administrative burden, costs practices money,

contributes to burnout, and its survey materials indicate some practices have staff working exclusively on PA. ([American Medical Association](#))

Why this is better than “save the world” projects

A disaster platform or climate platform may be more noble, but those usually involve:

- harder sales,
- public-sector procurement,
- lower willingness to pay at the start,
- and longer trust-building cycles.

A healthcare admin AI product sits inside a giant market with immediate pain and clearer economic value. That usually makes it easier to monetize earlier. The fact that CMS is actively creating rules and pilots around prior authorization technology tells you this is not imaginary demand. ([Centers for Medicare & Medicaid Services](#))

Why it is strong for an AI course

This is actually a very good AI project because it uses:

- NLP on messy clinical text,
- document understanding on insurer policy PDFs,
- information extraction,
- reasoning over rules,
- risk scoring,
- retrieval,
- and human-in-the-loop decision support.

And again, your a16z notes basically describe this category: messy enterprise data, claims/billing dispute adjudication, weak APIs, fragmented workflows, and AI products that become core infrastructure in regulated industries.

What the course-project version should be

Do **not** build the whole healthcare company.

Build the course version as:

"AI Assistant for Prior Authorization and Appeal Generation"

Core features:

- upload clinical note + payer policy + denial letter
- extract diagnosis, treatment, patient facts, and required criteria
- check policy match
- flag missing documentation
- output approval likelihood
- generate a draft prior-auth summary
- generate a draft appeal letter if denied
- show explanation: "why this may be denied"

That is enough to be impressive and still feasible.

What AI models/components you would use

Use a hybrid pipeline:

- document parser for PDFs
- medical/clinical NER
- retrieval over payer policy documents
- rules + reasoning layer
- classification model for denial-risk prediction
- LLM for draft generation
- confidence and explanation layer

This is the right mix because pure LLM-only systems are too fragile for regulated workflows.

The hard problems attached to it

This is important, because these are exactly the things that make the project serious:

1. Messy documents

Policies are in PDFs, notes are unstructured, denial letters vary by payer.

2. Explainability

You cannot just say "model says denied." You need traceable reasons.

3. Compliance and trust

In production, this would need strong review and privacy controls.

4. Payer variation

Different insurers have different rules and documentation expectations.

5. Human-in-the-loop design

Your system should assist staff, not autonomously approve care.

Those are not weaknesses. They are what make it portfolio-worthy.

How you would eventually make money from it

Sell it B2B:

- clinics
- specialty practices
- hospital revenue-cycle teams
- billing companies
- prior-auth outsourcing firms

Monetization models:

- per seat
- per provider

- per submission
- percentage of recovered revenue
- enterprise annual contract

The best version is probably: **base SaaS fee + usage fee + premium appeal automation.**

The best positioning

Do not pitch it as:

| “AI for healthcare”

Pitch it as:

| **“We reduce prior-auth delays and recover denied revenue.”**

That sells much better.

Name ideas

- AppealIQ
- PriorPilot
- AuthFlow AI
- DenialSense
- ClaimLift

My blunt recommendation

If you want the **best chance of building something that is both course-worthy and monetizable**, build:

PriorPilot — AI copilot for prior authorization, denial prediction, and appeal drafting

It has:

- a huge budget environment,
- a painful workflow,
- strong AI depth,
- a believable business model,
- and it matches the exact future-facing themes in your a16z material. (Centers for Medicare & Medicaid Services)

The runner-up would be an **AI permit/compliance copilot for construction/solar projects**, because permitting and soft costs are also expensive bottlenecks, but healthcare has the cleaner “people will pay now” story. (The Department of Energy's Energy.gov)

I can turn this into a full project brief with problem statement, objectives, architecture, datasets, AI pipeline, evaluation metrics, and MVP features.